

Salesperson Performance Analysis

Fair evaluation of salespeople using sales, profit, growth and average order value

Prepared by: Sneha | Data Analytics Track Intern, Veda Technology | Tools: SQL (MySQL), Excel, Power BI | Dataset: Superstore (2014-2017)

1. Executive Summary

This project ranks the four Superstore salespeople (regional managers) on **six KPIs** instead of revenue alone, and then adjusts for **territory differences** (number of states, customer base and discounting). Main findings:

- **Anna Andreadi (West) ranks #1** in both the absolute and the territory-fair model: highest sales (USD 725K), highest profit (USD 108K), best margin (14.9%) and lowest average discount (10.9%).
- **Chuck Magee (East) is a clear #2** with USD 679K sales and the best average order value (USD 484), but about one fifth of his sales are made at discounts above 20%, and Ohio and Pennsylvania lose USD 32.5K together.
- **Revenue alone is misleading:** Kelly Williams (Central) sells USD 110K more than Cassandra Brandow (South) yet earns USD 7K less profit. Central has a 7.9% margin because of heavy discounting (24% average) and losses in Texas (-USD 25.7K) and Illinois (-USD 12.6K).
- **Discounts above 20% are the biggest profit leak:** across the company they lost USD 135K, against a total profit of USD 286K.
- Ranking is stable: all three models/weightings give the same order - Anna, Chuck, Cassandra, Kelly.

2. Objective, Data and Approach

Objective: evaluate salesperson performance fairly. **Data:** Superstore orders, 9,994 order lines, 5,009 orders, 2,501 region-customer pairs, 49 states, Jan 2014 - Dec 2017; no missing values or duplicate rows were found.

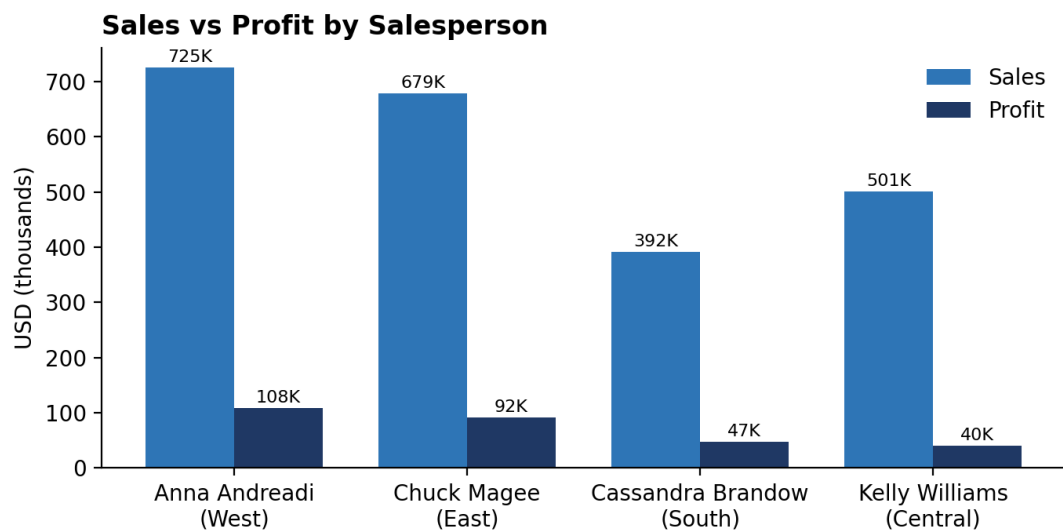
Important assumption: the Superstore file has no salesperson column. Following the standard Superstore "People" table, each salesperson is the Regional Manager of a region: West - Anna Andreadi, East - Chuck Magee, Central - Kelly Williams, South - Cassandra Brandow. Because one person owns one region, individual skill and territory quality cannot be fully separated (see Limitations).

Workflow: (1) data checks and salesperson mapping, (2) KPI calculation in SQL, (3) verification and scorecard in Excel (live formulas), (4) interactive dashboard in Power BI, (5) insights and recommendations.

3. KPI Comparison

KPI	Anna (West)	Chuck (East)	Cassandra (South)	Kelly (Central)
Total sales (USD)	725,458	678,781	391,722	501,240
Total profit (USD)	108,418	91,523	46,749	39,706
Profit margin	14.9%	13.5%	11.9%	7.9%
Orders	1,611	1,401	822	1,175
Avg order value (USD)	450.32	484.50	476.55	426.59
Sales growth 2017 vs 2016	+33.4%	+17.9%	+31.3%	-0.2%
Sales CAGR 2014-17	19.1%	18.3%	5.8%	12.3%
Avg discount	10.9%	14.5%	14.7%	24.0%
Loss-making order lines	9.9%	19.4%	16.0%	31.9%
States covered	11	14	11	13
Sales per state (USD)	65,951	48,484	35,611	38,557
Profit per customer (USD)	158.04	135.79	91.31	63.13

AOV = total sales / distinct orders. CAGR = (Sales 2017 / Sales 2014)^(1/3) - 1. Customers are counted per region.

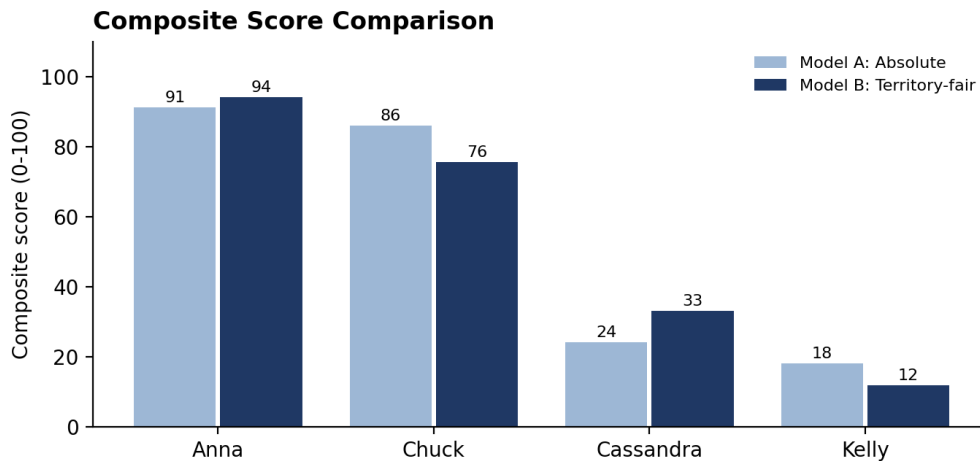


4. Ranking Method and Result

KPIs have different units, so each KPI is scaled 0-100 across the four people (min-max) and combined with weights. Two models were built:

- **Model A - Absolute:** sales 25%, profit 25%, margin 15%, growth (CAGR) 20%, average order value 15%.
- **Model B - Territory-fair (recommended):** sales per state 20%, profit per customer 20%, margin 20%, growth 20%, average order value 10%, discount discipline 10% (lower average discount scores higher).

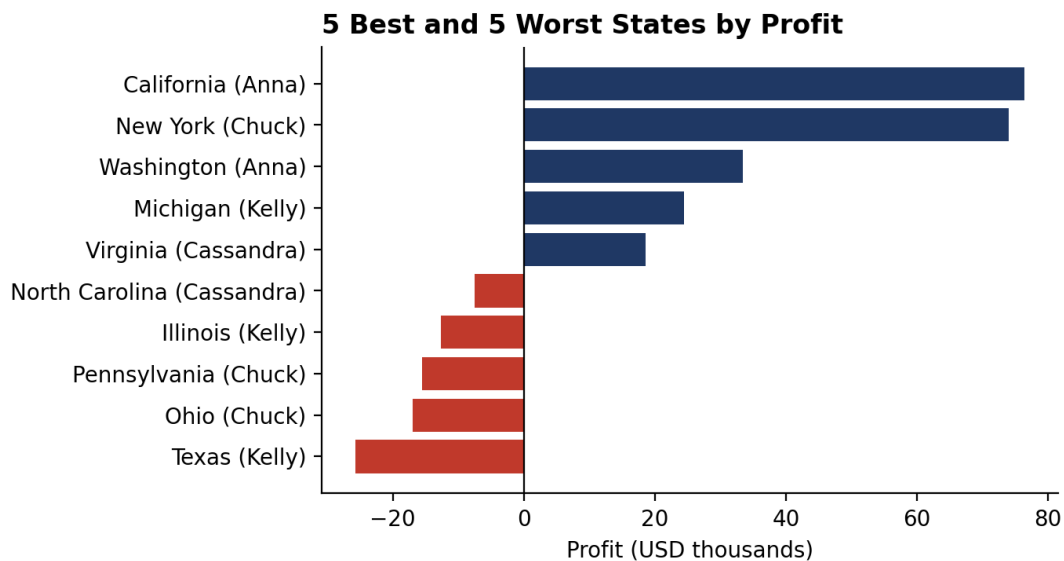
Rank	Model A (absolute)	Score	Model B (territory-fair)	Score
1	Anna Andreadi (West)	91.1	Anna Andreadi (West)	94.1
2	Chuck Magee (East)	86.0	Chuck Magee (East)	75.6
3	Cassandra Brandow (South)	24.1	Cassandra Brandow (South)	33.1
4	Kelly Williams (Central)	18.0	Kelly Williams (Central)	11.7



Weights are a judgement call and can be changed in the Excel Scorecard sheet; the order of the four people did not change in the tests done here. With only four people, the gaps between ranks matter more than the exact scores.

5. Territory Differences

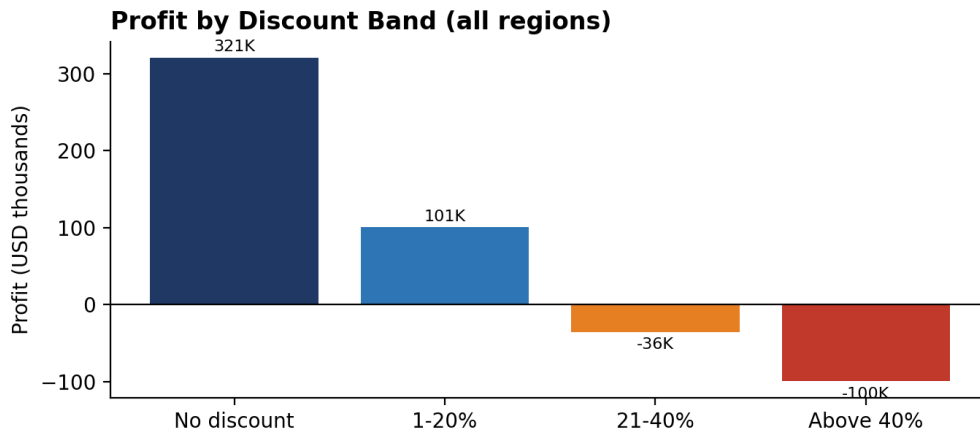
Territories are not equal. West has the highest sales per state (USD 65.9K) while South has the lowest (USD 35.6K), so raw totals favour people who happen to have bigger or richer markets. East covers 14 states, South and West only 11. Model B therefore compares performance **per state and per customer**.



Ten of the 49 states are loss-making: Texas (-25.7K), Ohio (-17.0K), Pennsylvania (-15.6K), Illinois (-12.6K), North Carolina (-7.5K), Colorado (-6.5K), Tennessee (-5.3K), Arizona (-3.4K), Florida (-3.4K) and Oregon (-1.2K). The largest profit pools are California (+76.4K) and New York (+74.0K). A large share of the gap between salespeople is explained by a handful of states.

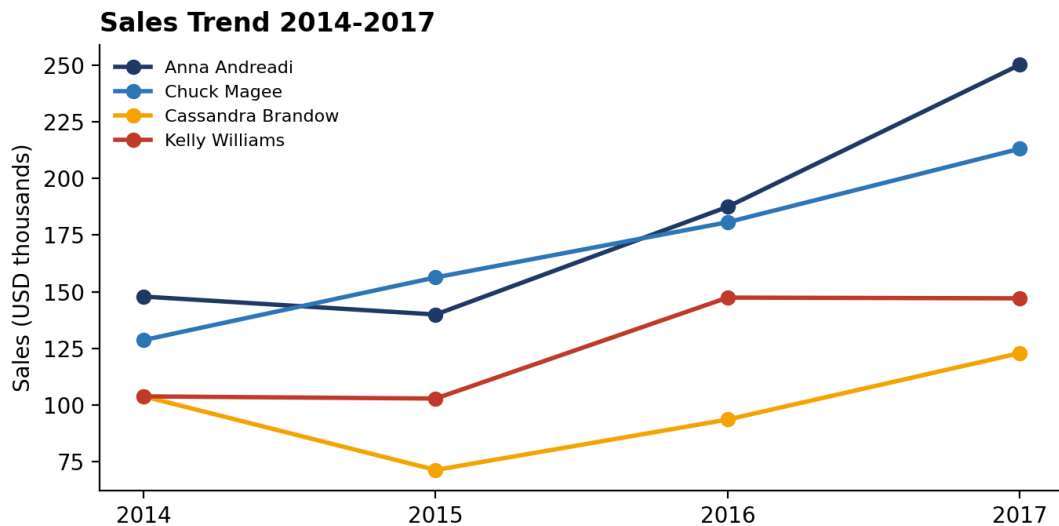
6. Discount Discipline

Profit falls sharply as discounts rise: lines with no discount earned USD 321K, lines at 1-20% earned USD 101K, while 21-40% lost USD 36K and above 40% lost USD 100K.

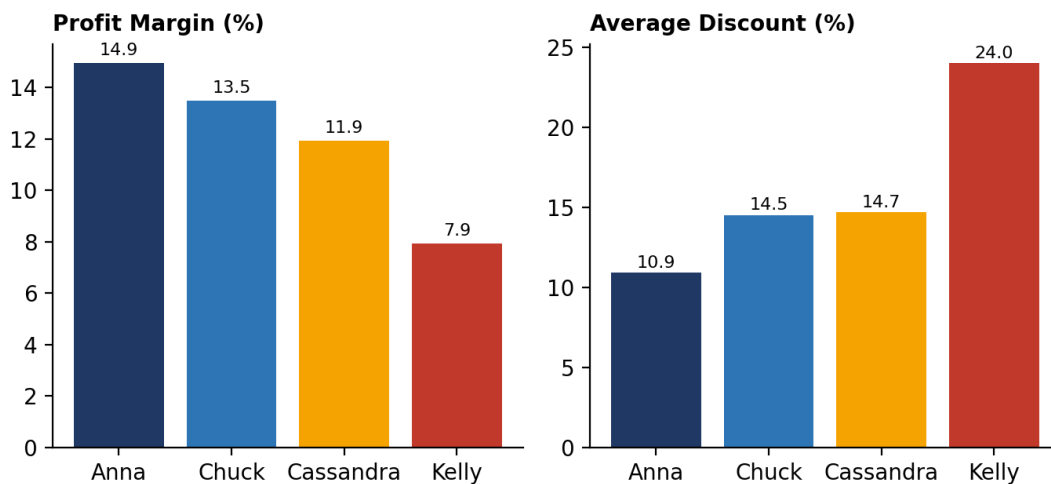


Share of each person's sales sold at more than 20% discount: Anna 2.4%, South 18.1%, East 21.5%, Kelly 25.8%. Anna's lead in margin is mostly a discount-policy story rather than a volume story.

7. Growth and Trend



West and East grew steadily. South dropped 31% in 2015 and then recovered with about +31% in both 2016 and 2017 - the best current momentum after West. Central stalled in 2017 (-0.2%) after a strong 2016 jump. Because of the 2015 dip, South's 3-year CAGR (5.8%) understates its recent recovery, so both CAGR and latest-year growth should be watched.



8. Recommendations

Company level

- Introduce a discount approval rule: anything above 20% needs manager sign-off. Lines above 20% lost USD 135K; even recovering part of that is worth more than most growth targets (this is an upper bound, because some volume would be lost).
- Set targets per state (potential-based) instead of one common revenue target, and review salespeople on margin and profit per customer along with sales.
- Run a review of the loss-making states (Texas, Ohio, Pennsylvania, Illinois, North Carolina) - check product mix, especially Furniture and Tables/Bookcases, and discount levels.

Per salesperson

- **Anna Andreadi (West):** best practice example - share her discount discipline. Opportunity: Furniture margin is only 4.6%; Colorado and Arizona are loss-making.
- **Chuck Magee (East):** strongest average order value and New York is a profit engine. Priority: fix Ohio and Pennsylvania and reduce deep discounts (21.5% of sales above 20% discount).
- **Cassandra Brandow (South):** good margin on small base, momentum improving. Priority: grow customer base (512 customers, lowest) and protect against another 2015-type dip; look at North Carolina and Tennessee.
- **Kelly Williams (Central):** highest priority for coaching. Cut discounts (24% average, 6% of sales above 40% discount), and fix Texas and Illinois, which together lose USD 38K. Central's Furniture business loses money overall.

9. Interview Questions - Short Answers

Why is revenue alone insufficient? It ignores cost of selling. Central out-sells South by USD 110K but makes less profit because of discounting; revenue also ignores growth, order size and customer base.

How can territory affect comparison? Territories differ in number of states, customers, market size and local pricing pressure. Comparing raw totals rewards the bigger market; per-state and per-customer metrics, and discount discipline, give a fairer view.

10. Limitations and Next Steps

- Only four people, one per region: skill and territory cannot be fully separated. Rep-level data (accounts, quota, headcount, tenure) would allow a real adjustment.
- Salesperson mapping is an assumption based on the standard Superstore people table.
- Scorecard weights are subjective; sensitivity checking is possible in the Excel Scorecard sheet.
- Data covers 2014-2017 only; no cost-to-serve, returns or target data.
- Next steps: add quota vs actual, returns data, and a forecast of 2018 per region.

11. Dashboard

The Power BI dashboard (Salesperson Performance) is attached after this page. It contains KPI cards, the leaderboard, sales vs profit by salesperson, yearly trend, state-level territory map/bar and discount impact, with slicers for Year and Category.

Files: Salesperson_Performance_Analysis.xlsx, salesperson_performance.sql, Salesperson_Performance.pbix (GitHub repository).